

PROBLEMS OF SYSTEMS ANALYSIS OF COMPUTER AIDED CIRCUIT DESIGN WITH DISPLAY UNIT

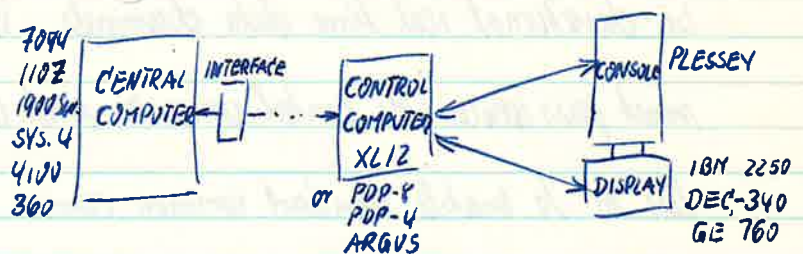
It is assumed that the system works on a shared memory basis of Control Computer and Central Computer. In the design of a display system for CAD the following features should receive attention:

- Preceding the design stage of the system a thorough analysis of the hardware and software should be undertaken.
- Hardware design should not limit programmer's flexibility

Hardware and software can be grouped into:

Hardware

- Control Computer
- Display Unit
- Central Computer
- Interface



SYSTEM BLOCK DIAGRAM

Software

- Softw. for Central C.
 - display
 - console
 - pre-processing.
- Softw. for Control C.
 - analysis
 - processing.

Systems Hardware

1.) Control Computer

This computer could be a XL12 with 8K to 16K 24 bits words which provides:

- a) System control (executive program)
- b) System memory
- c) Calculation of subroutines
- d) Graphics handling routines
- X e) Circuit modelling routines

The coupling to the other elements of the system can be carried out by three bi-directional real time data channels. The amount and rate of data which must pass between the control computer and central computer have to be sufficiently low to enable standard common carrier communication facilities to be used, e.g. not more than 2400 Baud.

Peripheral equipment should include a tape punch and read.

2) Display unit

This could be a digital controlled CRT display with 1024 by 1024 points representing an image of 10" square with alphanumeric and graphic capability.

Depending on complexity of circuits a vector length of 127 dots seemed to be sufficient and the generation time per point should be about 1 to 2 μ s.

The phosphorescence time should reduce flickering but not restrain dynamic changes in display. — how about steered beams?

The display unit shares the control computers memory. The number of units can be augmented if required. A great number of displays simultaneously in operation insures more economical use of central computer time.

Console

The communication devices for the operation of the system should contain:

Man to Machine:

- 1.) Keyboard for manual input and overall job control
- 2.) Light pen
- 3.) Rand tablet

Machine to Man

- 1.) CRT
- 2.) Lights under program control, mounted in the switches
- 3.) Small noise making device, operated by relays under program control

3.) Central Computer

The user is primarily concerned with the software facilities on the central computer. But the hardware needed for circuit analysis programs on a time shared basis should contain:

- 1.) 32K core storage 48 to 64 bit word length
- 2.) Input/output buffer channels
- 3.) 10 magnetic tape units
- 4.) Card reader and punch
- 5.) Line printer
- 6.) Paper tape punch and reader

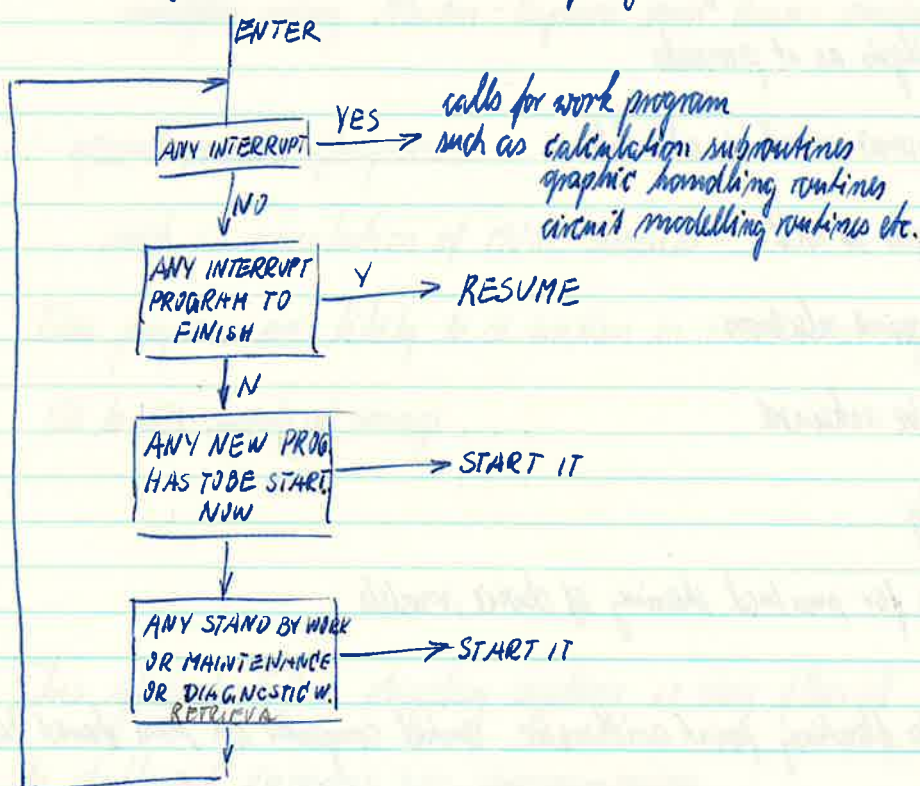
and a variety of programming languages.

The interface transfers data between the control computer and central computer. On the control computer side, the interface accesses memory via the data interrupt system and on the central computer side it is connected to a pair of input/output buffer channels.

Systems Software

1.) For Control Computer

a) EXECUTIVE is an interrupt-activated display system monitor which depends on periodic interrupts to initiate regular checks on the system status. This program supervises and handles interrupts, pushbuttons, lights, knobs, and the display.



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prog. { b) Graphic handling routines } relocatable subroutines which
c) Circuit modelling routines } reside in memory only when loaded

All programs are written in machine-language. It should be investigated whether the 24 bit word length of the XL12 requires programming in word pairs or whether individual instruction without concern about the modes of adjacent instructions are possible.

2) For Central Computer

Time-sharing environment and special requirements for Display Units impose special programming requirements upon analysis and systems monitor program (EXECUTIVE) within time-shared facilities.

If the analysis time on a problem is long and the number of simultaneous users large the following features are necessary for an efficient system:

a) Monitor of the analysis as it proceeds

b) Computational sequence must be alterable

X c) Stopping of analysis at will

d) Pseudo-random point selection

e) All data should be retained

X f) Queuing priority

X g) Library facilities for mutual sharing of device models

A 32K 48 bit words for floating point arithmetic central computer on time shared basis could fulfill these requirements.

X As it is not practical to interrupt the central computer during the execution of another customer's job, the average job execution time of the computer should not exceed 2 min.

On the other hand the core storage requirements for a circuit analysis computation does hardly permit access for other users to the central computer. Therefore further investigations should resolve the efficient usage of the idle time of the computer during the circuit designers contemplation time of the problem.

The EXECUTIVE program can be supplied of the manufacturer but will be very generalized, so many users write their own EXECUTIVES.

Other points for the investigations are:

a) further analysis and development of existing analysis programs.

This could include nonlinear extensions of ECAP for d-c and transient analysis using Newton-Raphson and Quasi-analytic methods,

b) special purpose programs

such as simulation of MOST-Circuits

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These programs are likely to be written in FORTRAN or Algol and require approx. 10K to 30K words of storage.

This general CAD display system is very flexible and can be adopted to different purposes via programming.